

JANE NJERI NJUGUNA

Data Scientist | Data Analyst | BI Analyst

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PROFESSIONAL SUMMARY

Data Scientist with experience delivering machine learning solutions, business intelligence dashboards, statistical analysis, and AI-powered applications across healthcare, agriculture, education, and retail domains. Skilled in Python, SQL, Power BI, PostgreSQL, machine learning, natural language processing, geospatial analytics, and predictive modeling. Experienced in transforming complex datasets into actionable insights through exploratory analysis, predictive analytics, dashboard development, reporting automation, and business intelligence solutions. Delivered 30+ dashboards and analytics solutions supporting operational performance, KPI monitoring, decision-making, and strategic planning.

TECHNICAL SKILLS

Programming & Databases: Python, SQL, PostgreSQL, MySQL

Analytics & Statistics: Exploratory Data Analysis (EDA), Statistical Analysis, Hypothesis Testing, Feature Engineering, Predictive Modeling, Forecasting, A/B Testing, Data Validation & Quality Assurance, Quantitative Analysis, Data Storytelling

Business Intelligence: Power BI, Tableau, Looker Studio, Excel, KPI Development, Dashboard Design, Reporting Automation, Business Analytics

Machine Learning & AI: Scikit-learn, LightGBM, XGBoost, Optuna, TensorFlow, PyTorch, Classification & Regression Modeling, Model Evaluation & Cross-Validation, Natural Language Processing (NLP), Transformers, Hugging Face, LLM Applications

Geospatial AI & Remote Sensing: Sentinel-2 Imagery, Remote Sensing Analytics, Raster Processing, Spatial Feature Engineering, Satellite Image Analysis, Geospatial Machine Learning

Tools & Platforms: Git, GitHub, Streamlit, AWS (S3), Jupyter Notebook, Google Colab, VS Code, Jira

PROFESSIONAL EXPERIENCE

Data Analyst | Kurasa Africa

Jun 2025 – Sep 2025

- Developed SQL reporting workflows and analytical solutions using PostgreSQL to support educational performance monitoring and operational decision-making.
- Designed and maintained interactive dashboards in Power BI and Looker Studio for student engagement, examination outcomes, and institutional reporting.
- Cleaned, transformed, and validated large-scale academic datasets, improving reporting accuracy and data reliability.
- Conducted exploratory data analysis to identify learning trends, performance gaps, and operational insights.
- Automated reporting processes and dashboard updates, reducing manual reporting effort and improving accessibility of insights.
- Contributed to the design and delivery of dashboards and reporting solutions supporting educational analytics, KPI monitoring, and institutional decision-making.

Data Analyst | Kiondo Market

Jul 2023 – Dec 2023

- Analyzed sales, inventory, and operational datasets across 500+ products using SQL, Excel, and Power BI.
- Developed KPI dashboards and business intelligence reports to support revenue monitoring, inventory optimization, and operational decision-making.
- Conducted customer segmentation, trend analysis, and profitability assessments to identify growth opportunities and business drivers.
- Automated recurring reporting workflows and strengthened data validation processes.
- Delivered actionable recommendations supporting pricing strategies, inventory planning, and performance improvement initiatives.
- Provided insights that improved visibility into sales performance, product demand, and operational efficiency.

KEY PROJECTS

Climate Health Risk Prediction | [GitHub](#)

Python, Scikit-learn, Pandas

Competition Ranking: 7th Place out of 99 Participants

- Developed a machine learning model to predict health risk levels using climate and environmental datasets.
- Performed feature engineering, exploratory data analysis, and predictive modeling to improve model performance.
- Applied classification algorithms and evaluation techniques to enhance predictive accuracy and model generalization.
- Generated interpretable insights supporting public health decision-making and early risk detection.
- Identified key environmental predictors influencing health outcomes.

Sentinel2 Multi-Temporal Crop Classification | [Github](#)

Python, LightGBM, Optuna, Rasterio, Scikit-learn | Zindi Competition

Award: 2nd Place – Top Female

- Developed a pixel-level crop classification model to distinguish cocoa, rubber, and oil palm plantations using multi-temporal Sentinel-2 satellite imagery from Côte d'Ivoire.
- Engineered advanced spectral, spatial, and seasonal features including NDVI, EVI, SAVI, NDWI, NDRE, texture metrics, and temporal crop-growth indicators.
- Built a LightGBM ensemble pipeline with Optuna hyperparameter optimization and GroupKFold cross-validation to prevent spatial leakage.
- Applied probability-weighted ensembling and post-processing techniques to improve model robustness and classification performance.
- Achieved a Macro F1 Score of 0.8663 with strong class-level precision and recall.

Medical Appointment No-Show Analysis Using EDA and Statistical Testing

Python, Pandas, Seaborn

- Analyzed 110,000+ healthcare appointment records to identify drivers of patient no-shows.
- Conducted exploratory data analysis and applied Mann-Whitney U and Chi-Square statistical tests.
- Identified waiting time as the strongest predictor of appointment non-attendance.
- Discovered that teenage patients exhibited the highest no-show rates compared to other age groups.
- Delivered actionable recommendations for scheduling optimization and patient engagement strategies.

ACHIEVEMENTS

- 2nd Place – Top Female, Côte d'Ivoire Crop Classification Challenge
- Top 10 Competition Finisher, Climate Health Risk Prediction Challenge (7th Place / 99 Participants)

EDUCATION

Diploma in Data Science & Analytics - Distinction (GPA: 3.72)

Zetech University

Jan 2024 – Nov 2025

Relevant Coursework: Machine Learning Models, Predictive Analytics, Python for Data Science, Artificial Intelligence, Data Mining & Warehousing, Data Analytics & Visualization

Nanodegree in Data Science

Moringa School

Jul 2022 – Feb 2023

BSc. Community Health and Development

Catholic University of Eastern Africa

Nov 2020 – Apr 2022

LANGUAGES

English (Fluent) **Swahili** (Fluent)